



Precalculus Honors

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–12 · FULL YEAR · HONORS

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in lesson order, activities, and examples from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Precalculus Honors extends a scholar's understanding of functions to the mathematics that calculus will assume. Right-triangle trigonometry grows into unit-circle trigonometry and the trigonometric functions that model periodic phenomena; conic sections appear as fundamental geometric forms; complex numbers and vectors get full treatment in the coordinate plane; and parametric and polar representations extend what a relation in the plane can be. Arithmetic and geometric sequences and series round out the year. VR and AI enhancements support visualizing periodic functions, transformations, and complex relationships. Throughout, scholars build precision in symbolic and graphical reasoning and care in understanding what these structures reveal about patterns in the world. Prerequisite: Algebra 2.

THE YEAR AT A GLANCE

1 Functions and Their Behavior (foundation)

Extending and deepening the function toolkit — analysis, transformations, and graphical reasoning with rigor.

3 Conic Sections (major unit)

Parabolas, ellipses, and hyperbolas as fundamental geometric forms.

5 Parametric and Polar Representations (major unit)

New ways to describe relations in the plane, extending what graphs can say.

2 Trigonometry (major unit)

From right triangles to the unit circle: trigonometric functions and the periodic phenomena they model.

4 Complex Numbers and Vectors (major unit)

Representing and operating with complex numbers and vectors in the coordinate plane.

6 Sequences and Series (major unit)

Arithmetic and geometric sequences and series, analyzed and applied.

The Honors difference. Increased pace and depth throughout, with rigor in analyzing function behavior and geometric properties — this is the direct preparation for Calculus.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- Visualization work with VR and AI tools brings periodic functions and transformations to life; quizzes and tests close each unit.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A graphing or scientific calculator — trigonometry and function analysis lean on it.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.